Introduction of Computer

A **computer** is an electronic device that accepts input, processes it according to a set of instructions (program), produces output, and can store data for future use.

It is a versatile machine capable of performing a wide range of tasks quickly and accurately.

Characteristics of Computer System

Computers possess several key characteristics that make them powerful tools:

- **Speed** — Computers perform operations at extremely high speeds, processing millions or
- billions of instructions per second.
- **Accuracy** — They produce highly precise results with minimal errors
- (errors usually occur due
- to human mistakes in input or programming).
- **Diligence** — Computers can work continuously for long hours without tiredness, b;’’[,,,,,,,78ioredom, or
- loss of concentration.
- **Versatility** — They can perform different types of tasks (e.g., calculations, graphics, multimedia,
- communication) with the same efficiency.
- **Reliability** — Consistent results are produced for the same input every time.
- **Storage Capacity** — Computers can store massive amounts of data and retrieve it instantly.
- **Automation** — Once instructions are given, computers execute tasks automatically without
- human intervention.

These features make computers indispensable in modern life.

Basic Applications of Computers

Computers are used in almost every field today. Some basic applications include:

- **Education** — Online classes, e-learning, research, and student record management.
- **Business & Commerce** — Accounting, inventory management, online marketing, banking
- (ATMs, transactions), and e-commerce.
- **Healthcare** — Patient records, monitoring vital signs, medical imaging, and robotic surgeries.
- **Entertainment** — Gaming, movies, music streaming, and social media.
- **Science & Engineering** — Simulations, research, CAD (design), and weather forecasting.
- **Defense & Government** — Missile tracking, data analysis, record-keeping, and surveillance.
- **Communication** — Email, video conferencing, and social networking.
- **Home** — Bill payments, online shopping, and home automation.

Computers have revolutionized how we work, learn, and live.

Major Components of a Computer System

Processing Unit (CPU - Central Processing Unit)

The **CPU** is the brain of the computer. It performs all calculations, logical operations, and

controls other components. It includes:

- **Arithmetic Logic Unit (ALU)** — Performs mathematical and logical operations.
- **Control Unit (CU)** — Directs the flow of data and instructions.

Keyboard, Mouse, and VDU

- **Keyboard** — Primary input device for typing text, numbers, and commands.

- **Mouse** — Pointing device used to move the cursor, select items, and interact with the
- graphical interface.
- **VDU (Visual Display Unit)** → Also called **Monitor** — Main output device that displays
- text, images, videos, and results visually.

Other Input Devices

Besides keyboard and mouse, common input devices include:

- Scanner → Converts printed documents/images into digital format.
- Microphone → For voice input and recording.
- Joystick → Used in gaming.
- Webcam → Captures video/images.
- Touchscreen → Allows direct input via finger touch.

Other Output Devices

Besides monitor, common output devices include:

- Printer → Produces hard copies (text/images) on paper.
- Speakers/Headphones → Output sound and audio.
- Plotter → Prints large-scale drawings/maps.
- Projector → Displays output on a large screen.

Computer Memory

Memory stores data and instructions. It is divided into:

- **Primary Memory (Main/Volatile Memory)**
    - **RAM (Random Access Memory)** → Temporary, fast, read/write memory; data is
    - lost when power is off.

- ◦ **ROM (Read Only Memory)** → Permanent; stores firmware/boot instructions; data is not
  - lost on power off.
- **Secondary Memory (Auxiliary/Non-Volatile Memory)**
  - Hard Disk Drive (HDD), Solid State Drive (SSD), Pen Drive, CD/DVD, External drives →
  - Large, permanent storage; data remains even when power is off.

Concept of Hardware and Software

Hardware

Hardware refers to the **physical, tangible** components of a computer that you can touch, such as:

- CPU, monitor, keyboard, mouse, motherboard, RAM, hard disk, etc.

Software

Software is the set of **instructions/programs** that tell hardware what to do. It is intangible.

Software is divided into:

- **System Software**

  Controls and manages hardware operations.

  Examples: Operating Systems (Windows, Linux, macOS), device drivers, utilities.

- **Application Software**

  Designed to perform specific user tasks.

  Examples: Microsoft Word (word processing), Excel (spreadsheets),

  browsers (Chrome), games, photo editors.

Programming Languages

These are used to write software. They include:

- **Low-level** — Machine language (binary 0s & 1s), Assembly language (close to hardware).
- **High-level** — Human-readable like C, Java, Python, JavaScript (easier to code and understand).

Representation of Data/Information

Computers understand only **binary** (machine language) — combinations of **0** and **1** (off/on states).

All data is converted into binary:

- Numbers → Binary numbers.
- Text → Using codes like **ASCII/Unicode** (each character gets a binary code, e.g., 'A' = 01000001).
- Images → Pixels represented as RGB binary values.
- Sound/Video → Sampled and converted to binary.

A group of 8 bits = **1 byte** (can represent 256 values).

Concept of Data Processing

Data processing is the transformation of **raw data** (facts/figures) into meaningful **information** through a series of operations.

Basic steps (cycle):

1. **Input** → Collect raw data.
2. **Processing** → Perform calculations, sorting, classification using CPU.
3. **Output** → Present results (screen, print, etc.).
4. **Storage** → Save data/information for future use.

Types include batch processing (large volumes at once), real-time processing (instant results), etc.

Computers excel at data processing due to their speed, accuracy, and automation capabilities.

This covers the fundamental concepts of computers as requested! Let me know if you'd like more details on any section. 😊